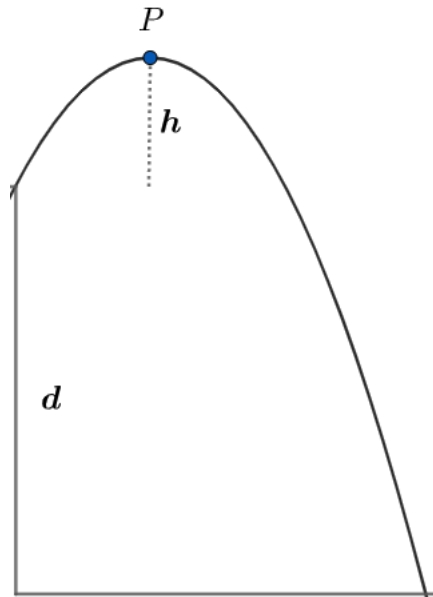


2021 F=ma Exam: Problem 11

Kevin S. Huang



From kinematics, the time t_1 it takes to reach the peak P is

$$\frac{1}{2}gt_1^2 = h$$

$$t_1 = \sqrt{\frac{2h}{g}}$$

The time t_2 it takes to reach the ground from the peak is

$$\frac{1}{2}gt_2^2 = h + d$$

$$t_2 = \sqrt{\frac{2(h + d)}{g}}$$

so the total time is

$$T = t_1 + t_2 = \sqrt{\frac{2h}{g}} + \sqrt{\frac{2(h + d)}{g}}$$

We see T increases with peak height h so the answer is D.